

Task 1: Fundamentals of Automobile Systems and Introduction to Electric Vehicles

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Introduction

In this task, I studied the basic systems of automobiles and learned about Electric Vehicles (EVs). I understood how conventional vehicles and electric vehicles work and the major differences between them.

Automobile Systems

Engine System

The engine is the main part of a vehicle. It converts the fuel energy into mechanical energy and helps the vehicle move. Petrol and diesel engines are commonly used in vehicles.

Transmission System

The transmission system transfers power from the engine to the wheels. It includes components such as clutch, gearbox, and differential. This system helps the vehicle run at different speeds.

Braking System

The braking system is used to stop or slow down the vehicle. Disc brakes and drum brakes are commonly used in automobiles. Brakes are important for vehicle safety.

Steering System

The steering system helps the driver control the direction of the vehicle. It allows the vehicle to turn left or right easily.

Suspension System

The suspension system reduces shocks and vibrations from rough roads. It improves comfort and stability while driving.

Electrical System

The electrical system consists of a battery, lights, horn, sensors, and wiring. It provides electrical power to different parts of the vehicle.

Electric Vehicle (EV) Components

Battery Pack

The battery pack stores electrical energy and supplies power to the vehicle.

Battery Management System (BMS)

The BMS monitors the battery and protects it from overcharging and overheating.

Inverter

The inverter converts DC power from the battery into AC power for the motor.

Electric Motor

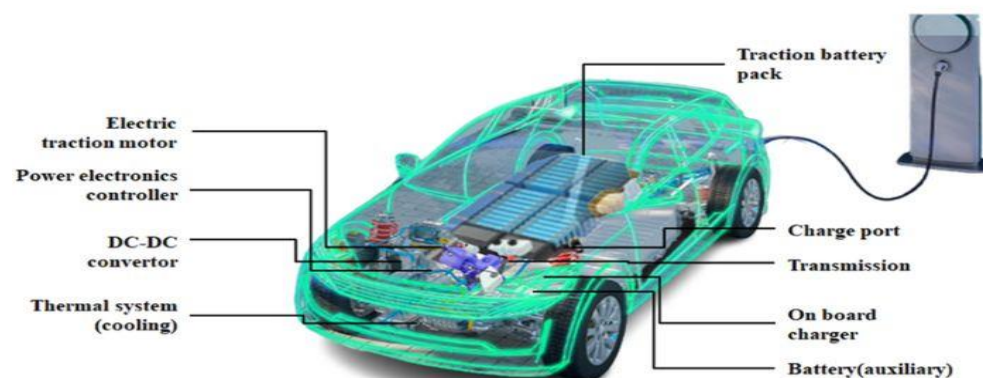
The motor converts electrical energy into mechanical energy to drive the wheels.

Charger

The charger is used to charge the battery from an external power source.

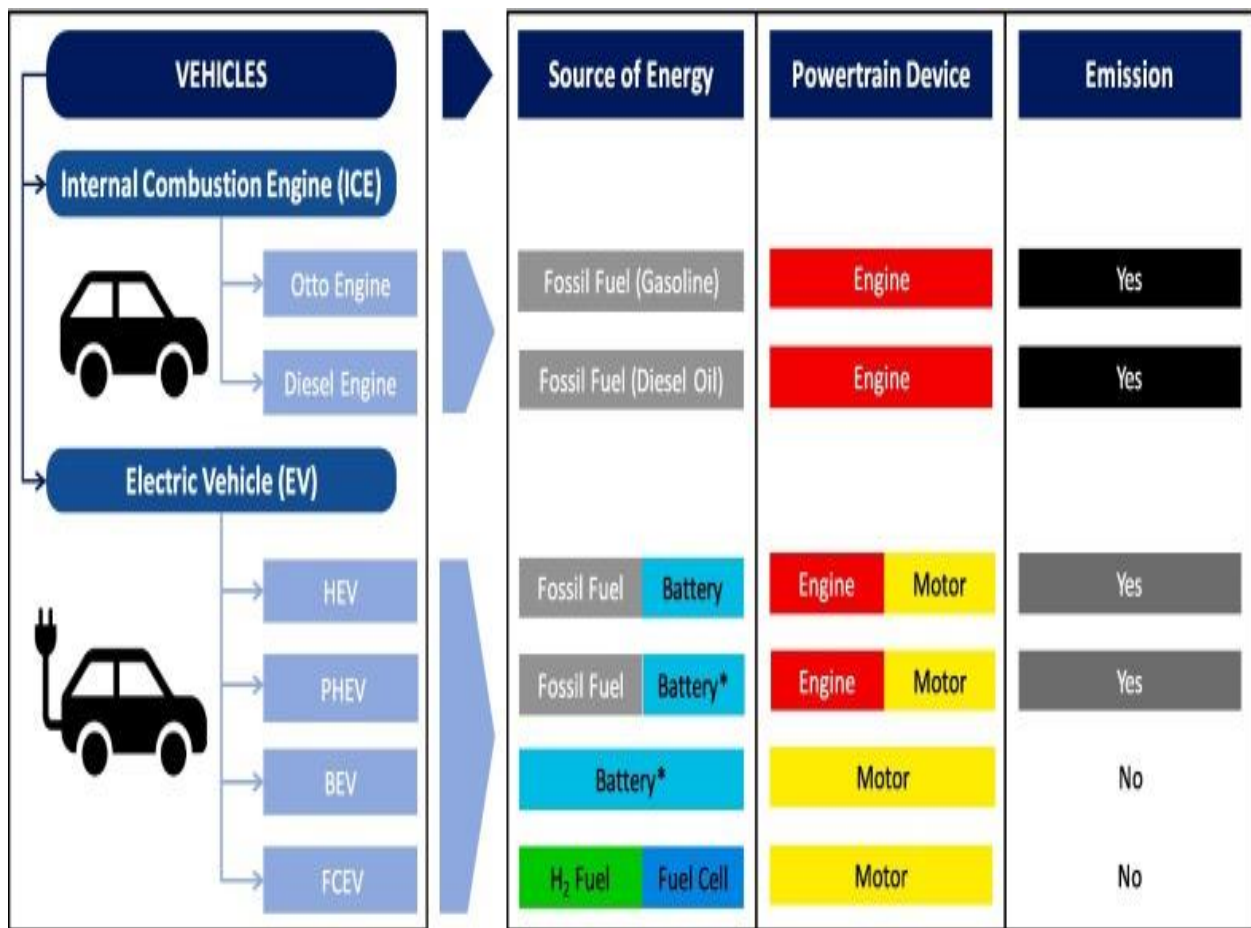
Working of an Electric Vehicle

1. The battery stores electrical energy.
2. The inverter receives power from the battery.
3. The inverter supplies power to the electric motor.
4. The motor rotates and produces mechanical power.
5. The wheels move and the vehicle runs.



Comparison Between ICE Vehicles and EVs

Feature	ICE Vehicle	EV
Fuel Source	Petrol/Diesel	Electricity
Pollution	More	Very Less
Noise	High	Low
Maintenance	High	Low
Efficiency	Less	More



*Completed with Plug In

Advantages of EVs

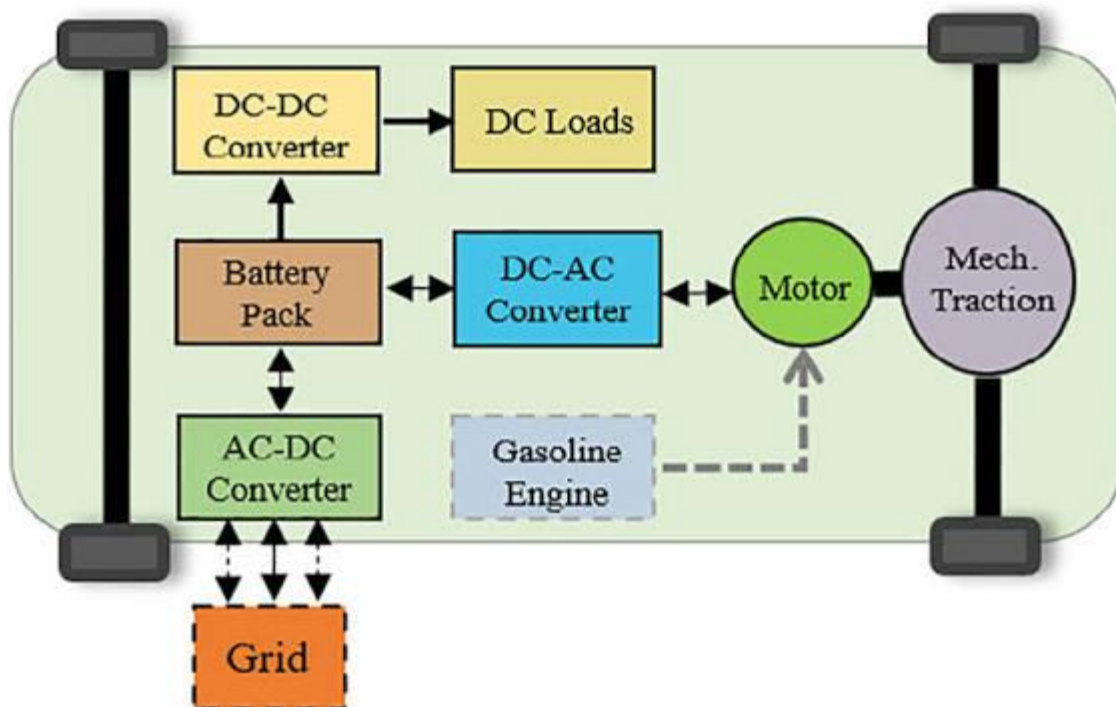
- Environment friendly

- Low maintenance cost
- Less noise pollution
- Better energy efficiency

Challenges of EVs

- Charging takes time
- Charging stations are limited in some places
- Battery cost is high
- Driving range is limited compared to fuel vehicles

EV Block Diagram



Conclusion

From this task, I learned about the basic automobile systems and the working of electric vehicles. I also understood the differences between conventional vehicles and EVs. Electric vehicles are becoming popular because they are environmentally friendly and have lower operating costs.